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# Credit Card Behaviour Score Predicting Default Probability

*Developing a Risk Management Framework for Credit Card Defaults*



Project By -

**Saket Mehla | Sahaj Yadav**

**Abstract:** This report presents an end-to-end credit risk modelling framework for predicting the probability of credit card default using anonymized behavioural, transaction and bureau attributes. The final solution combines leakage-safe feature discovery, extensive imbalance-learning experiments, five-fold out-of-fold ensemble optimization and business-oriented threshold analysis. The locked holdout achieved a ROC-AUC of 0.8574 and PR-AUC of 0.1112, while the highest-risk 5% of accounts contained approximately 39.8% of all defaults.

# Report Contents

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Description of Problem Statement</b>                                | <b>2</b>  |
| 1.1      | Key Objectives . . . . .                                               | 2         |
| 1.2      | Why the Problem is Different from Ordinary Classification . . . . .    | 2         |
| <b>2</b> | <b>Abstract of Proposed Solution</b>                                   | <b>3</b>  |
| <b>3</b> | <b>Dataset Overview</b>                                                | <b>4</b>  |
| 3.1      | Column Overview . . . . .                                              | 4         |
| 3.2      | Target Distribution . . . . .                                          | 4         |
| 3.3      | Data Integrity . . . . .                                               | 4         |
| 3.4      | Experimental Split . . . . .                                           | 5         |
| 3.5      | Evaluation Framework . . . . .                                         | 5         |
| <b>4</b> | <b>EDA with Important Graphs and Observations</b>                      | <b>6</b>  |
| 4.1      | Missingness Structure . . . . .                                        | 6         |
| 4.2      | Row-level Missingness . . . . .                                        | 7         |
| 4.3      | Feature-family Signal . . . . .                                        | 7         |
| 4.4      | Univariate Signal . . . . .                                            | 7         |
| 4.5      | Quantile Default-rate Profiles . . . . .                               | 8         |
| 4.6      | Information Value . . . . .                                            | 9         |
| 4.7      | Correlation and Redundancy . . . . .                                   | 10        |
| 4.8      | Development-versus-Scoring Drift . . . . .                             | 11        |
| <b>5</b> | <b>Missing Value Strategy and Data Quality</b>                         | <b>12</b> |
| 5.1      | Why High-missingness Features Were Not Automatically Dropped . . . . . | 12        |
| 5.2      | Imputation Philosophy . . . . .                                        | 12        |
| 5.3      | Leakage Prevention . . . . .                                           | 12        |
| 5.4      | Constant and Near-constant Features . . . . .                          | 12        |
| <b>6</b> | <b>Feature Engineering and Selection</b>                               | <b>13</b> |
| 6.1      | ANOVA F-test . . . . .                                                 | 13        |
| 6.2      | Mutual Information . . . . .                                           | 13        |
| 6.3      | Directional Single-feature ROC-AUC . . . . .                           | 13        |
| 6.4      | Kolmogorov-Smirnov Statistic . . . . .                                 | 13        |
| 6.5      | Selector Agreement . . . . .                                           | 14        |
| 6.6      | PCA as an Alternative Representation . . . . .                         | 14        |
| 6.7      | Representation Screening Results . . . . .                             | 15        |
| <b>7</b> | <b>Sampling Techniques and Imbalance Learning</b>                      | <b>16</b> |
| 7.1      | Nature of the Imbalance Problem . . . . .                              | 16        |
| 7.2      | Random Undersampling . . . . .                                         | 16        |
| 7.3      | Random Oversampling . . . . .                                          | 16        |
| 7.4      | SMOTE . . . . .                                                        | 16        |
| 7.5      | Borderline-SMOTE and ADASYN . . . . .                                  | 16        |

|           |                                                                 |           |
|-----------|-----------------------------------------------------------------|-----------|
| 7.6       | Balanced Ensembles . . . . .                                    | 16        |
| 7.7       | Comparison of Imbalance Treatments . . . . .                    | 17        |
| 7.8       | Cost-sensitive XGBoost . . . . .                                | 17        |
| 7.9       | Cost-sensitive CatBoost . . . . .                               | 18        |
| 7.10      | Hybrid Random Undersampling + Positive Sample Weights . . . . . | 18        |
| 7.11      | Conclusion from Imbalance Experiments . . . . .                 | 19        |
| <b>8</b>  | <b>Model Training</b>                                           | <b>20</b> |
| 8.1       | Logistic Regression . . . . .                                   | 20        |
| 8.2       | XGBoost . . . . .                                               | 20        |
| 8.3       | LightGBM . . . . .                                              | 20        |
| 8.4       | CatBoost . . . . .                                              | 20        |
| 8.5       | Feature-path Screening Summary . . . . .                        | 20        |
| <b>9</b>  | <b>Out-of-Fold Ensemble Optimization</b>                        | <b>22</b> |
| 9.1       | Why a Second Validation Stage Was Needed . . . . .              | 22        |
| 9.2       | Final OOF Feature Path . . . . .                                | 22        |
| 9.3       | Base Models . . . . .                                           | 22        |
| 9.4       | Weight Search . . . . .                                         | 22        |
| 9.5       | Threshold Selection from OOF Predictions . . . . .              | 23        |
| <b>10</b> | <b>Final Locked Holdout Evaluation</b>                          | <b>24</b> |
| 10.1      | Ranking Performance . . . . .                                   | 24        |
| 10.2      | Threshold Policies . . . . .                                    | 24        |
| 10.3      | Calibration . . . . .                                           | 25        |
| 10.4      | Capture and Lift . . . . .                                      | 25        |
| 10.5      | Risk Deciles . . . . .                                          | 26        |
| <b>11</b> | <b>Model Interpretation and Final Feature Importance</b>        | <b>27</b> |
| 11.1      | Weighted Ensemble Importance . . . . .                          | 27        |
| 11.2      | Interpretation Limitations . . . . .                            | 27        |
| <b>12</b> | <b>Summary Table of Experiments and Decisions</b>               | <b>28</b> |
| <b>13</b> | <b>Final Model, Scoring Output, and Production Logic</b>        | <b>30</b> |
| 13.1      | Final Ensemble . . . . .                                        | 30        |
| 13.2      | Final Performance Summary . . . . .                             | 30        |
| 13.3      | Production Refit . . . . .                                      | 30        |
| 13.4      | Final Output File . . . . .                                     | 30        |
| 13.5      | Operational Use . . . . .                                       | 30        |
| <b>14</b> | <b>Key Findings, Limitations, and Future Work</b>               | <b>32</b> |
| 14.1      | Key Findings . . . . .                                          | 32        |
| 14.2      | Limitations . . . . .                                           | 32        |
| 14.3      | Future Work . . . . .                                           | 32        |
| <b>15</b> | <b>References</b>                                               | <b>34</b> |

# 1 Description of Problem Statement

Credit card issuers continuously make decisions under uncertainty: whether an account is likely to remain healthy, whether a customer should receive additional exposure, and which accounts deserve early risk intervention. A behaviour score converts a customer’s recent behavioural, on-us, transaction and bureau information into a risk ranking that can support such decisions.

The objective of this project is to predict the probability of credit card default using a large anonymized tabular dataset. The labelled development file contains 96,806 accounts and 1,214 predictors, while a separate scoring file contains 41,792 accounts for which default probabilities must be generated. The target variable, `bad_flag`, equals 1 for a default and 0 otherwise.

The central technical difficulty is severe class imbalance. Only 1,372 of the 96,806 labelled accounts default, corresponding to a prevalence of approximately 1.417%. Thus, there is roughly one default for every 69.6 non-defaults. A classifier that predicts every account as non-default would already obtain approximately 98.6% accuracy, despite having zero practical ability to identify risk. Consequently, the project cannot be evaluated through accuracy alone.

## 1.1 Key Objectives

- **Data Exploration and Quality Assessment:** understand the raw structure, missingness, sparsity, class imbalance and stability of the anonymized features.
- **Leakage-safe Feature Discovery:** identify informative variables using complementary statistical measures without allowing the final holdout to influence the selection process.
- **Imbalance Investigation:** systematically benchmark undersampling, oversampling, synthetic sampling, balanced ensembles and cost-sensitive learning rather than assuming a single treatment must work.
- **Model Development:** compare linear, gradient-boosting and ensemble approaches using metrics suitable for rare-event prediction.
- **Robust Validation:** move beyond one tuning split by using stratified out-of-fold predictions for final ensemble selection and threshold design.
- **Risk Ranking:** measure how effectively the model concentrates defaults in the highest-risk fractions of the portfolio through capture and lift.
- **Production Scoring:** train on all available labelled observations only after evaluation is complete and generate continuous default probabilities for the 41,792 scoring accounts.

## 1.2 Why the Problem is Different from Ordinary Classification

With prevalence  $\pi \approx 0.01417$ , even a small false-positive rate can produce many false positives because the negative class is very large. Precision therefore tends to be numerically modest even when ranking quality is useful. For this reason, the project emphasizes ROC-AUC, PR-AUC, precision, recall, F1, F2, KS, capture and lift. Threshold-dependent metrics are interpreted as operating policies rather than as properties of the probability model itself.

## 2 Abstract of Proposed Solution

The proposed solution follows an experiment-driven workflow rather than a predetermined modelling recipe. Every major modelling choice is treated as a hypothesis to be tested on data.

1. **Experimental Split:** the labelled file is divided into 70% training, 15% tuning and 15% locked holdout using stratification. The holdout is not used for feature selection, resampling decisions, model weighting, calibration choice or threshold search.
2. **Full-training EDA:** feature-level statistics are computed on the complete training split. Expensive nonlinear feature-ranking calculations use a stratified training-only sample, while final model fitting continues to use the complete relevant data.
3. **Multi-view Feature Ranking:** ANOVA F-statistic, Mutual Information, directional single-feature ROC-AUC and KS are used together. Their disagreement is treated as useful information, and a 300-feature candidate pool is created instead of assuming that a single statistical test is sufficient.
4. **Credit-risk Diagnostics:** quantile default-rate profiles, Weight of Evidence, Information Value, correlation structure and development-versus-scoring drift are examined.
5. **Representation Tournament:** supervised feature subsets are compared against PCA. PCA is retained only if it demonstrates superior predictive performance; in the executed experiments it does not.
6. **Model Screening:** Logistic Regression, XGBoost, LightGBM and CatBoost are compared across raw and selected feature paths.
7. **Imbalance Benchmarking:** random undersampling, random oversampling, SMOTE, Borderline-SMOTE, ADASYN, class weighting, Balanced Random Forest, EasyEnsemble and RUSBoost are tested empirically. Mild resampling occasionally helps, but aggressive balancing generally reduces precision-recall quality.
8. **Cost-sensitive Boosting:** moderate positive-class penalties are investigated. Unlike extreme balancing, moderate penalties improve minority sensitivity without destroying ranking quality.
9. **Hybrid Undersampling + Sample Weights:** random undersampling is also combined with positive-class sample weights. The best tuning-split combination (RUS to 5% positives + weight 5) initially improves PR-AUC to 0.1052, but falls to 0.0786 PR-AUC under five-fold OOF validation and is therefore rejected.
10. **Five-fold OOF Ensemble:** after broad experimentation, training and tuning data are merged into an 85% development pool. Three complementary base models generate stratified out-of-fold probabilities. Ensemble weights are then searched over the probability simplex; no manual weights are assumed beforehand.
11. **Locked Holdout Evaluation:** the selected OOF ensemble is refit on the development pool and evaluated exactly once on the locked holdout. Threshold policies and portfolio capture are then reported.
12. **Final Refit and Scoring:** after validation is complete, the selected models are refit on all 96,806 labelled accounts and used to generate default probabilities for all 41,792 scoring accounts.

The final ensemble consists of 10% ordinary XGBoost, 25% moderately cost-sensitive XGBoost and 65% CatBoost. On the locked holdout it obtains a ROC-AUC of 0.8574 and PR-AUC of 0.1112. At the precision-constrained operating policy, precision is 11.62%, recall is 32.04% and F1 is 17.05%. At a recall-sensitive F2 operating point, recall increases to 48.06%. In ranking terms, the top 5% of accounts contain 39.81% of all holdout defaults and the top 20% contain 76.21%.

### 3 Dataset Overview

#### 3.1 Column Overview

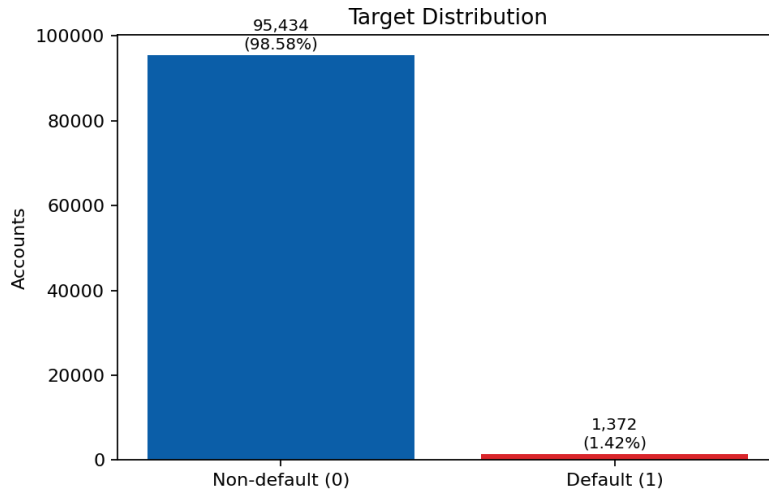
The labelled file contains 1,216 columns: the account identifier, the binary target and 1,214 anonymized predictors. The predictors can be grouped by their prefixes into four broad families.

**Table 1:** Feature-family overview

| Family         | Features | Mean Missing | Mean Univ. AUC |
|----------------|----------|--------------|----------------|
| Transaction    | 664      | 26.07%       | 0.5133         |
| Bureau         | 452      | 3.89%        | 0.5324         |
| Bureau Enquiry | 50       | 2.71%        | 0.5718         |
| On-us          | 48       | 16.94%       | 0.5638         |

The transaction family dominates raw dimensionality and is highly sparse and zero-heavy. Bureau-enquiry and on-us variables, despite being much smaller families, contain stronger average standalone discrimination. This does not imply that transaction variables are useless; their value may emerge through nonlinear interactions captured by tree ensembles.

#### 3.2 Target Distribution



**Figure 1:** Target distribution in the labelled development data

The positive class has a prevalence of only 1.417%. This immediately changes metric selection. The random PR-AUC baseline is approximately the positive prevalence, or 0.0142, whereas random ROC-AUC is 0.5. Thus, a PR-AUC of 0.10 is not directly comparable to an ROC-AUC of 0.85; each metric uses a different scale and answers a different question.

#### 3.3 Data Integrity

The account identifier is unique and the target contains no missing values. Two constant-like predictors are observed in the raw scan. Nineteen features have more than 50% missing values, but high missingness alone is not used as an automatic deletion rule because missingness itself can be predictive.

### 3.4 Experimental Split

**Table 2:** Initial stratified split

| Split   | Rows   | Defaults | Default Rate |
|---------|--------|----------|--------------|
| Train   | 67,764 | 960      | 1.417%       |
| Tuning  | 14,521 | 206      | 1.419%       |
| Holdout | 14,521 | 206      | 1.419%       |

The holdout remains locked until all feature-selection, imbalance-handling and ensemble decisions are made. After exploratory model development, train and tuning are merged into an 82,285-row development pool containing 1,166 defaults, and five-fold stratified out-of-fold modelling is used for final ensemble design.

### 3.5 Evaluation Framework

The project distinguishes between ranking metrics and operating-point metrics.

**ROC-AUC.** The probability that a randomly selected defaulter receives a higher score than a randomly selected non-defaulter.

**PR-AUC.** The average precision across recall levels. This is especially important in rare-event settings because precision directly reflects the proportion of true events among predicted positives [3].

**Precision and Recall.**

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \quad (1)$$

**F1 and F2.**

$$F_1 = 2 \frac{PR}{P + R}, \quad F_2 = 5 \frac{PR}{4P + R}. \quad (2)$$

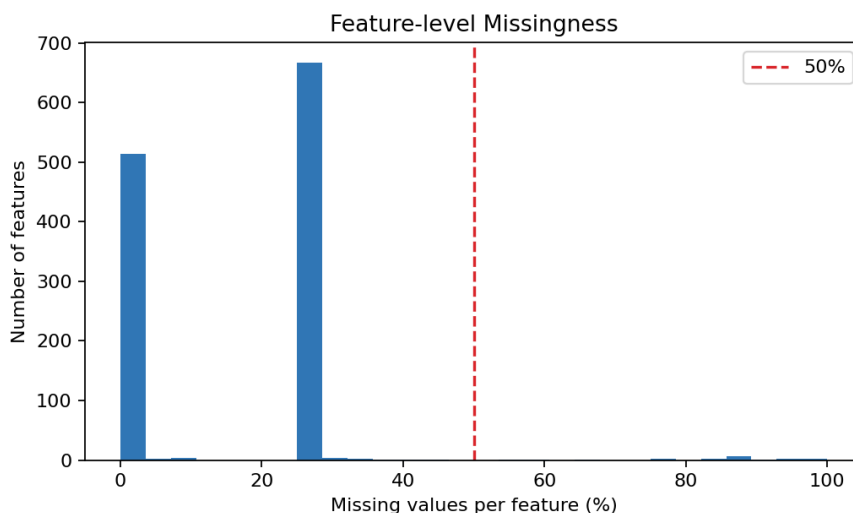
F2 is used as a recall-sensitive diagnostic because missing a default can be more costly than flagging an additional good account.

**Capture and Lift.** If the portfolio is sorted by predicted risk, Capture@ $k$  is the fraction of all defaults inside the top  $k$  fraction of accounts. Lift compares this capture with random selection:

$$\text{Lift@}k = \frac{\text{Capture@}k}{k}. \quad (3)$$

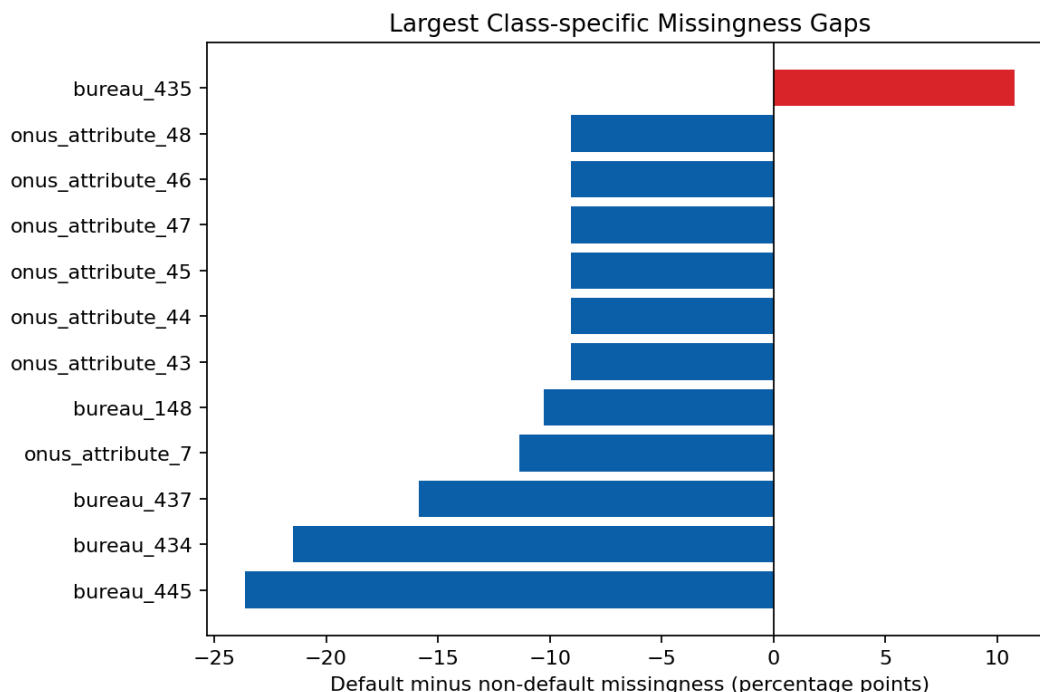
## 4 EDA with Important Graphs and Observations

### 4.1 Missingness Structure



**Figure 2:** Distribution of feature-level missingness

Nineteen features are more than 50% missing, with a small number exceeding 90% missingness. However, class-specific missingness reveals that the absence of a value can itself be informative. For example, `bureau_445` is missing for roughly 54.1% of non-defaults but only 30.5% of defaults, a gap of approximately 23.6 percentage points.



**Figure 3:** Largest class-specific missingness differences

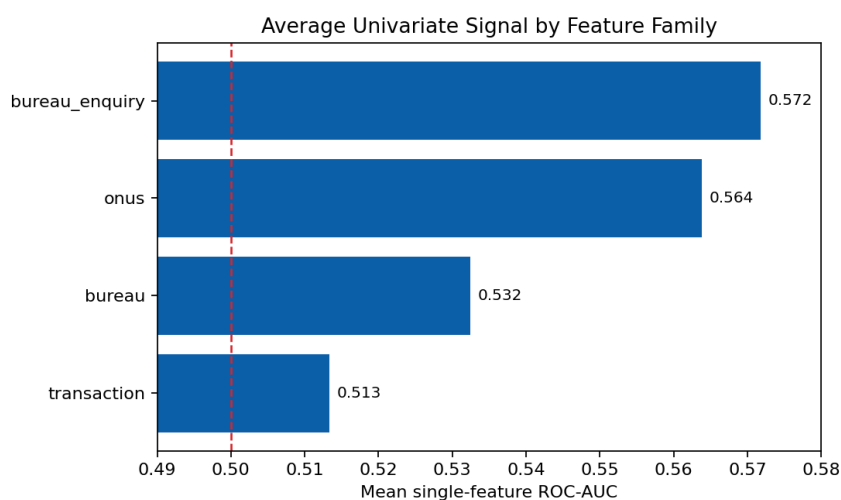
This observation argues against a blanket rule that removes all highly sparse variables. It also supports tree models that can often handle nonlinear missingness structure naturally or through explicit imputation plus indicators.

## 4.2 Row-level Missingness

For each account, row-level missingness is defined as the fraction of predictors that are missing. The mean missing share is approximately 16.50% for non-defaults and 15.36% for defaults. The median is much lower in both groups, around 1.9% and 1.7% respectively, indicating a heterogeneous population in which some rows are extremely sparse while many rows are nearly complete.

The aggregate row missingness difference is modest. The more informative effect is feature-specific: particular variables display much larger default/non-default missingness gaps than the overall row-level measure. Therefore, the model should not rely on a single “total missing count” as the primary missingness representation.

## 4.3 Feature-family Signal

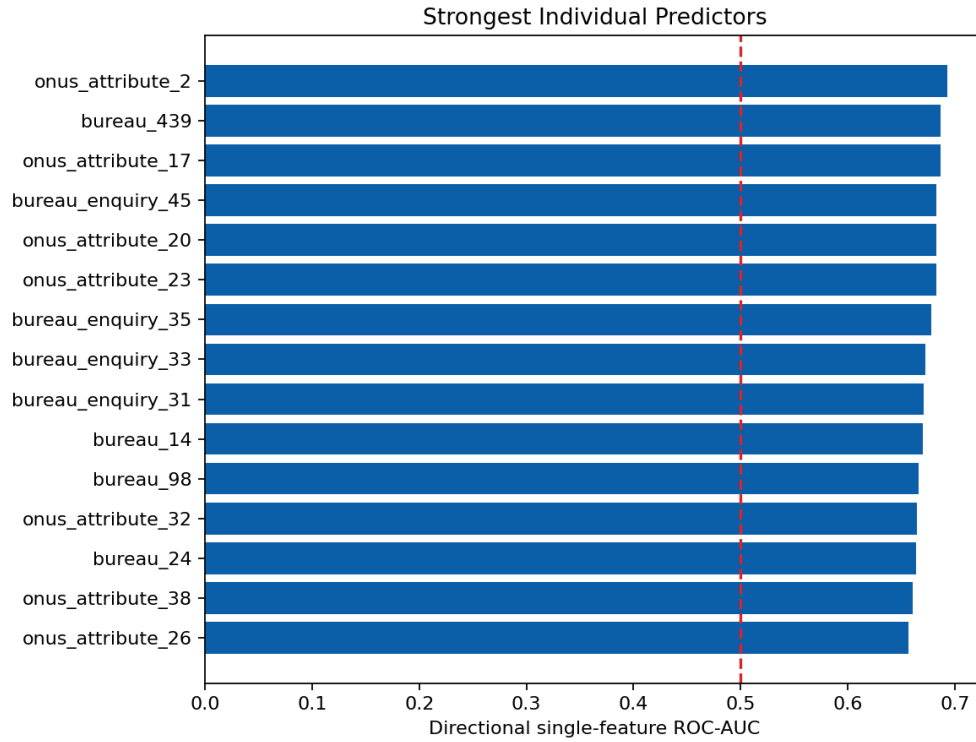


**Figure 4:** Average directional univariate ROC-AUC by feature family

On-us and bureau-enquiry variables exhibit stronger average individual predictive strength than the much larger transaction family. This result guides, but does not dictate, feature selection. Tree models may still extract interaction value from individually weak transaction attributes.

## 4.4 Univariate Signal

The strongest single variables include `onus_attribute_2`, `onus_attribute_17`, `onus_attribute_20`, `onus_attribute_23`, `bureau_439` and several bureau-enquiry variables.

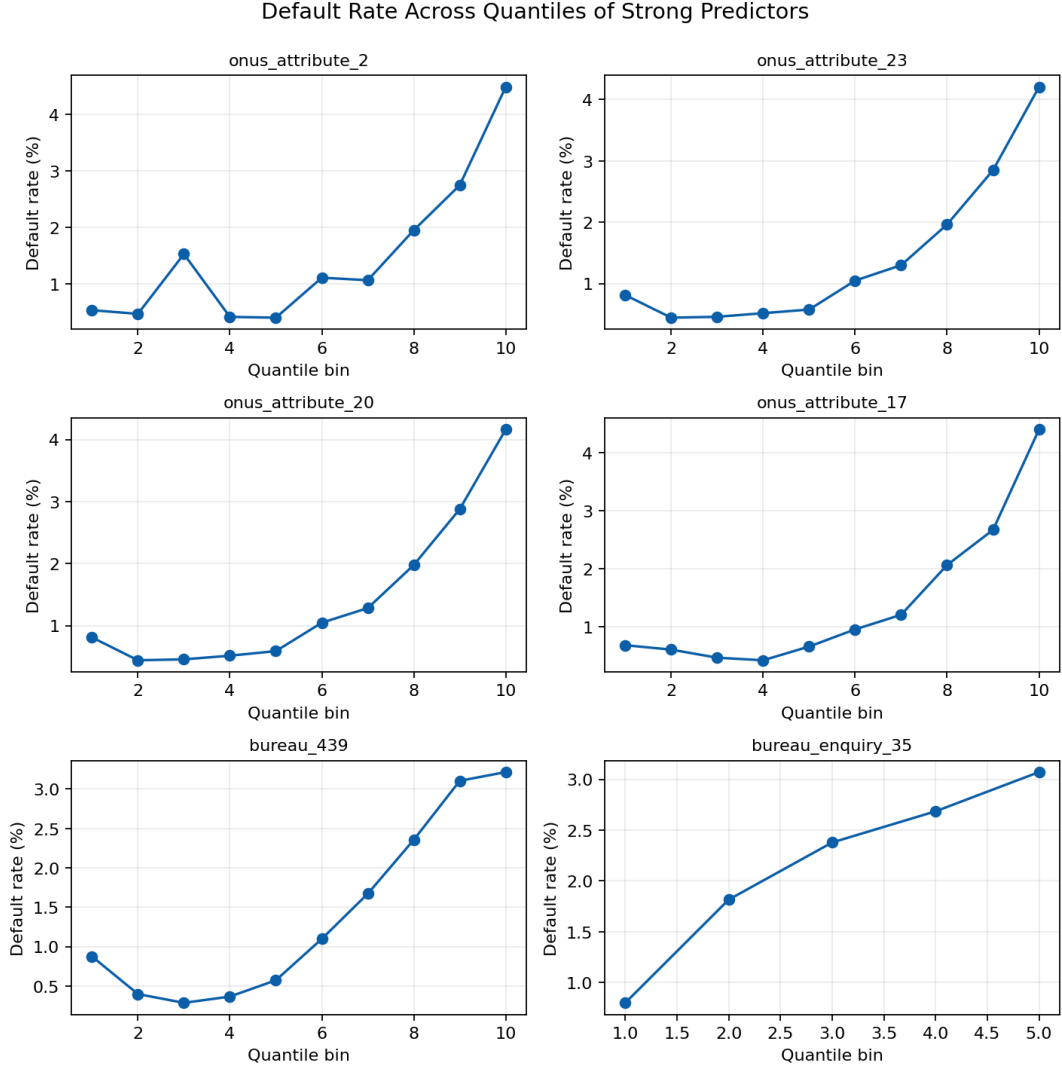


**Figure 5:** Top single-feature directional ROC-AUC values

A single-feature ROC-AUC close to 0.69 means that the feature alone has meaningful ranking information, but it is far from perfect separation. This is consistent with the final model: defaults and non-defaults overlap materially, so a multivariate nonlinear model is necessary.

#### 4.5 Quantile Default-rate Profiles

For the strongest predictors, values are binned into empirical quantiles and the observed default rate is computed within each bin.



**Figure 6:** Default rate across quantiles of selected strong predictors

These profiles reveal monotonic and threshold-like relationships that are not fully captured by mean differences alone. Such shapes provide a practical reason to prefer nonlinear boosting methods over relying solely on Logistic Regression.

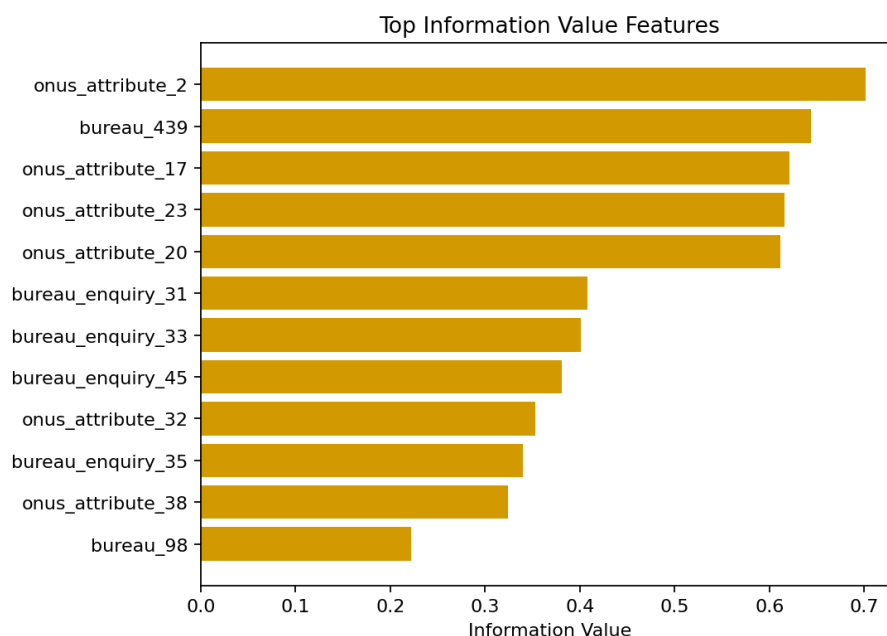
#### 4.6 Information Value

Weight of Evidence for a bin  $b$  is defined as

$$WOE_b = \ln \left( \frac{P(X \in b \mid Y = 1)}{P(X \in b \mid Y = 0)} \right), \quad (4)$$

and Information Value is

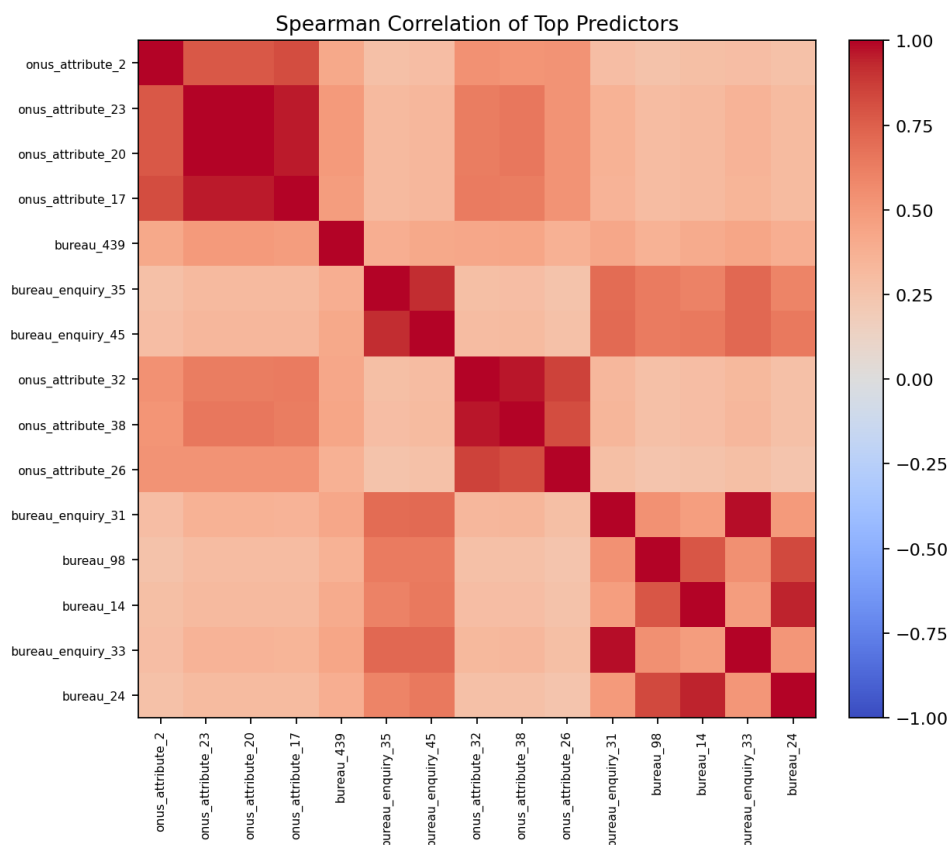
$$IV = \sum_b \left( P_b^{(1)} - P_b^{(0)} \right) WOE_b. \quad (5)$$



**Figure 7:** Information Value for leading risk predictors

The highest IV values are approximately 0.70 for `onus_attribute_2`, 0.64 for `bureau_439`, and above 0.61 for several related on-us attributes. IV is used as a credit-risk diagnostic rather than as the sole feature-selection rule.

## 4.7 Correlation and Redundancy

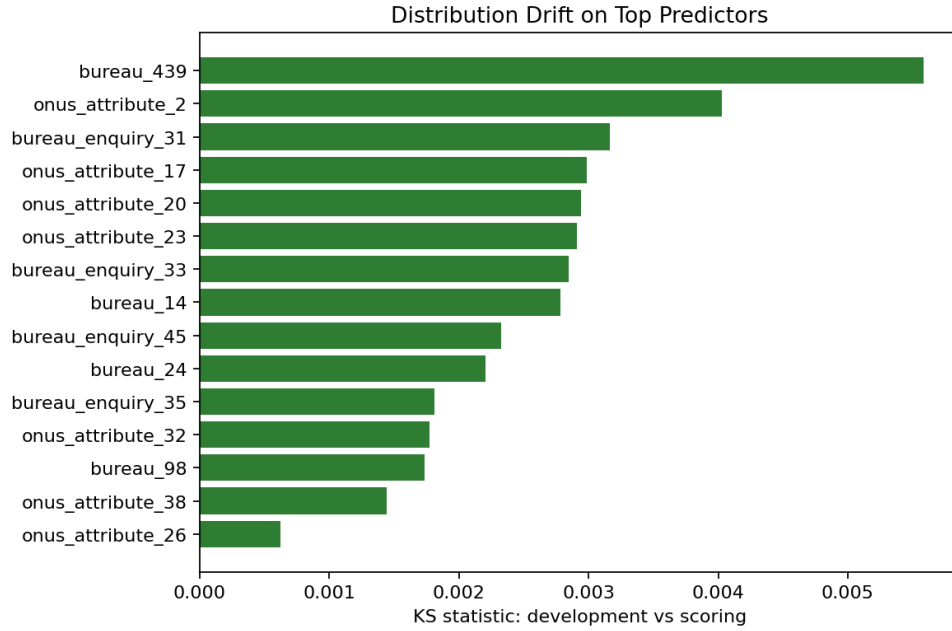


**Figure 8:** Spearman correlation among leading predictors

Strong redundancy exists among several on-us variables; for example, some pairs approach correlation of 1.0. This motivates testing PCA for linear representations. However, tree ensembles can use correlated variables without the same coefficient-instability concern as linear models, so correlation is not treated as an automatic deletion criterion.

#### 4.8 Development-versus-Scoring Drift

The top predictors are compared between labelled development data and the unlabeled scoring population using two-sample KS statistics and missingness differences.



**Figure 9:** Distribution drift of strongest predictors

The observed KS values for the strongest predictors are very small, typically below 0.006. No obvious large covariate shift is detected among these top variables. This does not prove that the future target distribution is identical, but it reduces concern that the most important raw predictors are distributed very differently in the scoring set.

## 5 Missing Value Strategy and Data Quality

### 5.1 Why High-missingness Features Were Not Automatically Dropped

A feature can be highly missing and still be useful if its missingness is associated with the target. The class-specific missingness analysis demonstrates exactly this behaviour for several bureau and on-us variables. Consequently, the pipeline avoids a naive “drop every feature above 50% missing” rule.

### 5.2 Imputation Philosophy

For representation methods and models that require a complete matrix, median imputation is used because it is robust to the severe skew and extreme values present in many features. Scaling is applied only where mathematically necessary, such as Logistic Regression and PCA. Tree-based methods are allowed to operate on the original feature scale.

More complex imputation procedures such as MICE or iterative regression can be attractive in principle, but with 1,214 predictors they greatly increase computational burden and risk of fitting intricate relationships that may not generalize. The final project prioritizes leakage-safe, reproducible preprocessing and relies on strong nonlinear models to absorb residual structure.

### 5.3 Leakage Prevention

All learned preprocessing objects are fitted only on the appropriate training portion. Feature discovery uses training labels, tuning is used for early model comparisons, and the holdout is excluded from modelling decisions. In the final OOF stage, each fold’s validation predictions are generated from models that were not trained on that fold.

This distinction is particularly important for PCA, imputation, resampling and calibration. Fitting such transformations on the complete labelled dataset before evaluation would allow information from future validation rows to influence the learned representation and would produce optimistic estimates.

### 5.4 Constant and Near-constant Features

Two constant-like columns are found in the full raw audit and carry no usable discrimination. Such features are removed from candidate consideration. Highly zero-inflated variables are not necessarily removed, because a rare non-zero event may still carry predictive signal.

## 6 Feature Engineering and Selection

### 6.1 ANOVA F-test

For a candidate feature  $X$ , ANOVA evaluates the ratio of between-class variation to within-class variation. In the binary case the statistic can be written conceptually as

$$F = \frac{\text{between-class variance}}{\text{within-class variance}}. \quad (6)$$

It is effective when default and non-default groups differ in mean, but it does not capture all nonlinear or non-monotonic associations.

### 6.2 Mutual Information

Mutual Information measures reduction in uncertainty of the target after observing a feature:

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}. \quad (7)$$

Its advantage is that it can detect nonlinear dependence that may have weak mean separation.

### 6.3 Directional Single-feature ROC-AUC

Each feature is treated as a one-dimensional risk score. If raw AUC is below 0.5, the feature may still be informative in the opposite direction; therefore the directional score uses

$$AUC_{dir} = \max(AUC, 1 - AUC). \quad (8)$$

This prevents a strongly inverse predictor from being incorrectly classified as useless.

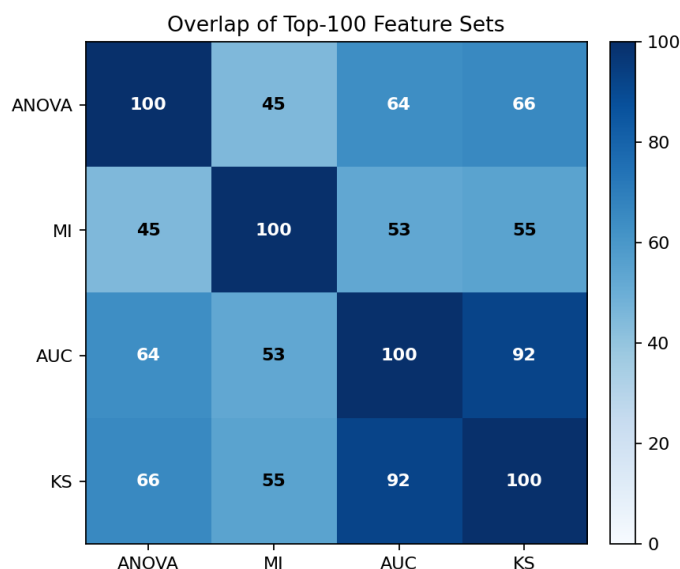
### 6.4 Kolmogorov-Smirnov Statistic

For a feature  $X$ , let  $F_1(x)$  be the empirical CDF among defaults and  $F_0(x)$  the empirical CDF among non-defaults. The KS statistic is

$$KS = \max_x |F_1(x) - F_0(x)|. \quad (9)$$

It measures the maximum vertical separation between the two class-conditional distributions.

## 6.5 Selector Agreement



**Figure 10:** Overlap between the top-100 sets of four feature-ranking methods

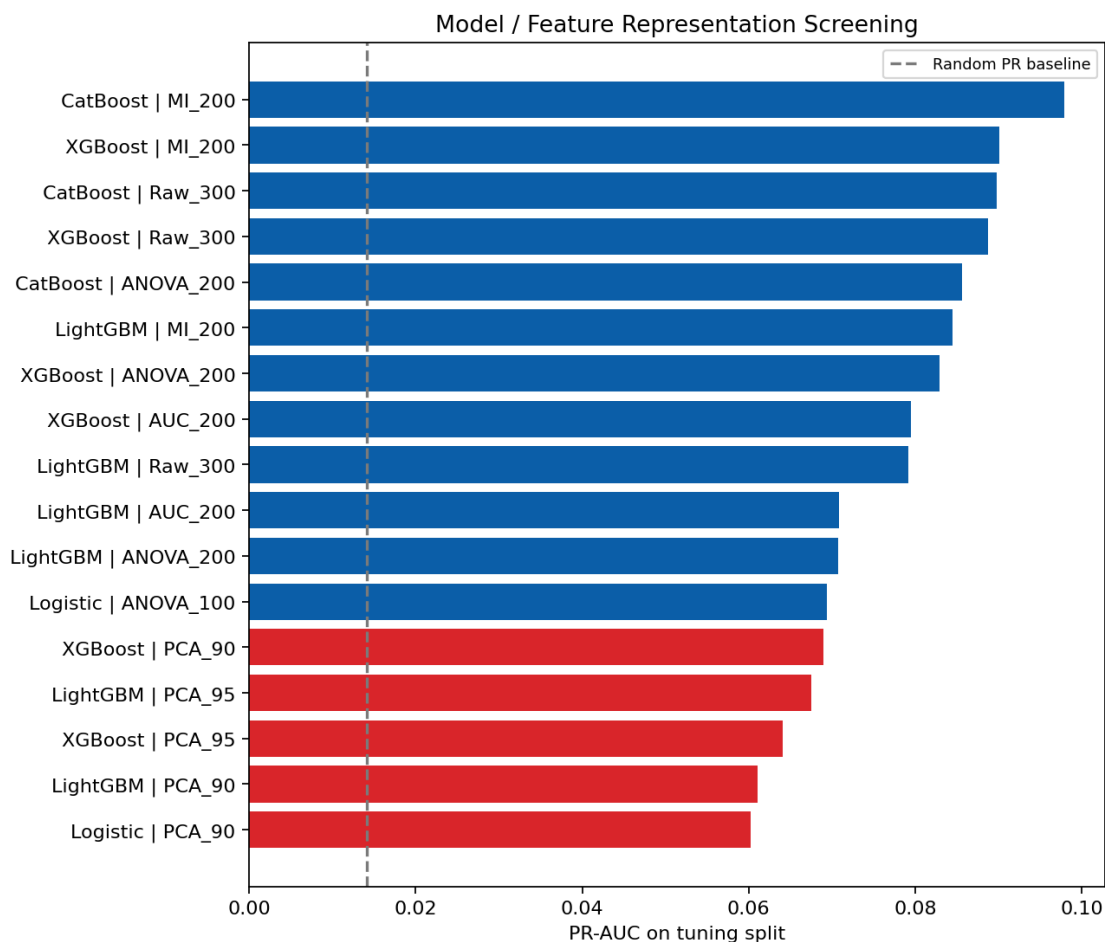
AUC and KS agree strongly, sharing 92 of their top 100 predictors. ANOVA and MI overlap on only 45 of their top 100, indicating that they detect meaningfully different signal. Rather than arbitrarily selecting one ranking method, the project averages ranks across ANOVA, MI, directional AUC and KS to form a broad 300-feature exploratory pool.

## 6.6 PCA as an Alternative Representation

Principal Component Analysis is explicitly tested rather than assumed to be appropriate or inappropriate. After median imputation and standardization of the candidate pool, 97 principal components are required to retain 90% of the variance and 130 components to retain 95%.

PCA optimizes variance of  $X$ , not discrimination with respect to  $Y$ . This distinction matters because a low-variance direction may still contain default signal. The empirical model comparison therefore determines whether PCA is retained.

## 6.7 Representation Screening Results



**Figure 11:** PR-AUC across model and feature-representation combinations

For Logistic Regression, ANOVA-100 obtains ROC-AUC 0.8014 and PR-AUC 0.0694, while PCA-90 obtains ROC-AUC 0.7941 and PR-AUC 0.0602. For boosted trees, PCA performs even worse: XGBoost PCA-90 obtains PR-AUC 0.0690 compared with 0.0902 for MI-200, while CatBoost MI-200 achieves the strongest early screening PR-AUC of 0.0979.

**Decision:** PCA is rejected for the final boosted-tree pipeline because supervised/raw feature representations consistently provide better tuning-set ranking. Mutual Information top 200 is retained for the final OOF stage because it performs strongly with nonlinear models, particularly CatBoost.

## 7 Sampling Techniques and Imbalance Learning

### 7.1 Nature of the Imbalance Problem

The data contains only 1,372 real defaults. Resampling can change the class proportions presented to the learner, but it cannot create genuinely new information about default behaviour. In a high-dimensional sparse space, synthetic interpolation can also create samples that do not perfectly reflect real customer structure. These concerns are consistent with the broader imbalanced-learning literature, which emphasizes that no resampling method dominates universally [2, 4].

### 7.2 Random Undersampling

Random undersampling reduces the majority class. Its advantage is computational simplicity and improved minority exposure, but its primary cost is discarded information. In the executed experiment, RUS at a 10% minority ratio obtains PR-AUC 0.0891, similar to the no-balancing XGBoost baseline of 0.0896. Because it removes many legitimate non-default observations without improving the key metric, it is rejected.

### 7.3 Random Oversampling

Random oversampling duplicates minority observations. At a 10% minority ratio it produces PR-AUC 0.0845, lower than the baseline. Duplicating rare events increases their influence but does not add behavioural diversity, and the executed result does not justify retaining it.

### 7.4 SMOTE

SMOTE synthesizes a minority sample along the line segment between a minority observation  $x_i$  and one of its minority neighbours  $x_{nn}$  [1]:

$$x_{new} = x_i + \lambda(x_{nn} - x_i), \quad \lambda \in [0, 1]. \quad (10)$$

Mild SMOTE to approximately a 5% positive rate improves XGBoost PR-AUC from 0.0896 to 0.0929 and ROC-AUC from 0.8076 to 0.8200. However, increasing the synthetic positive share to approximately 10% reduces PR-AUC to 0.0795. Thus, mild SMOTE is useful evidence that additional minority exposure can help, but stronger synthetic balancing degrades ranking.

### 7.5 Borderline-SMOTE and ADASYN

Borderline-SMOTE focuses synthetic samples near the decision boundary, while ADASYN places more synthetic examples around minority observations considered difficult to learn. In the executed runs, Borderline-SMOTE reaches PR-AUC between approximately 0.0727 and 0.0813. ADASYN reaches approximately 0.0775 to 0.0847. Neither beats the best cost-sensitive boosting candidates.

### 7.6 Balanced Ensembles

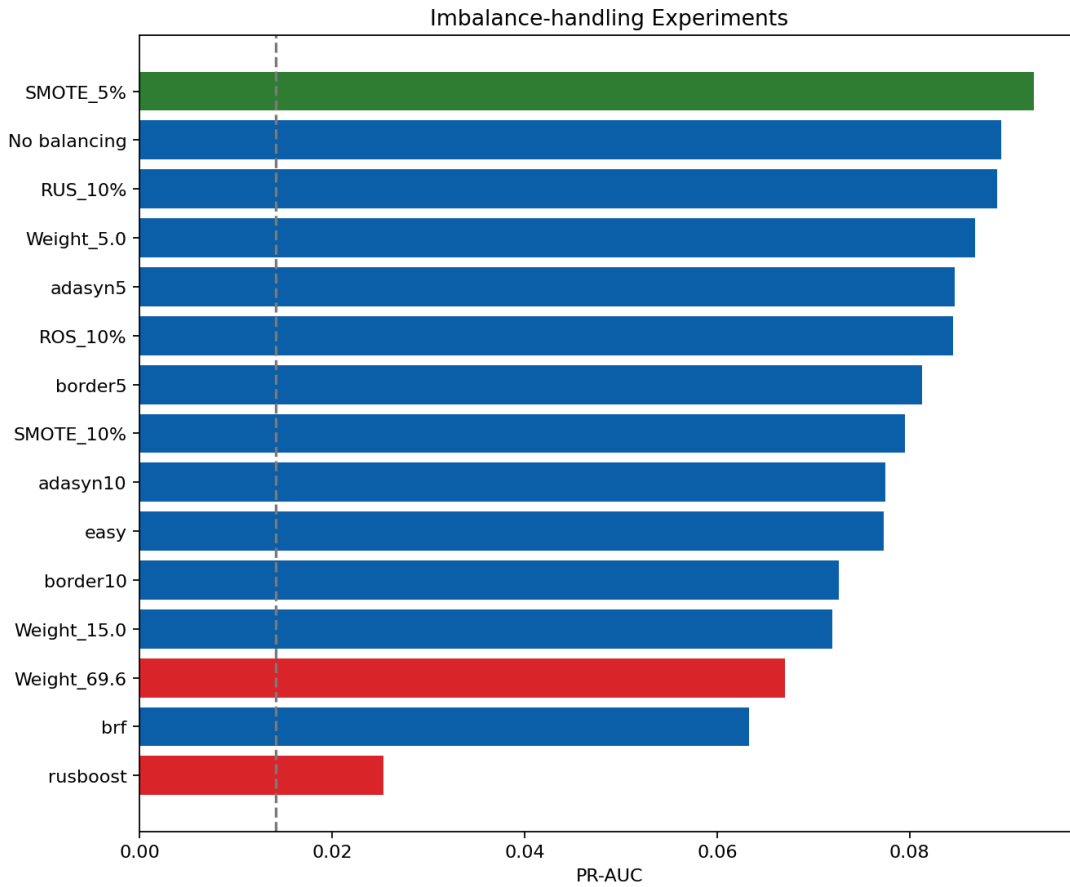
Balanced Random Forest, EasyEnsemble and RUSBoost are evaluated as dedicated imbalance-learning algorithms.

**Table 3:** Balanced ensemble results on the tuning split

| Model                  | PR-AUC | ROC-AUC | Max-F1 Recall | Max-F2 Recall |
|------------------------|--------|---------|---------------|---------------|
| Balanced Random Forest | 0.0633 | 0.8007  | 16.50%        | 36.89%        |
| EasyEnsemble           | 0.0773 | 0.7976  | 14.08%        | 41.75%        |
| RUSBoost               | 0.0254 | 0.6747  | 61.17%        | 61.17%        |

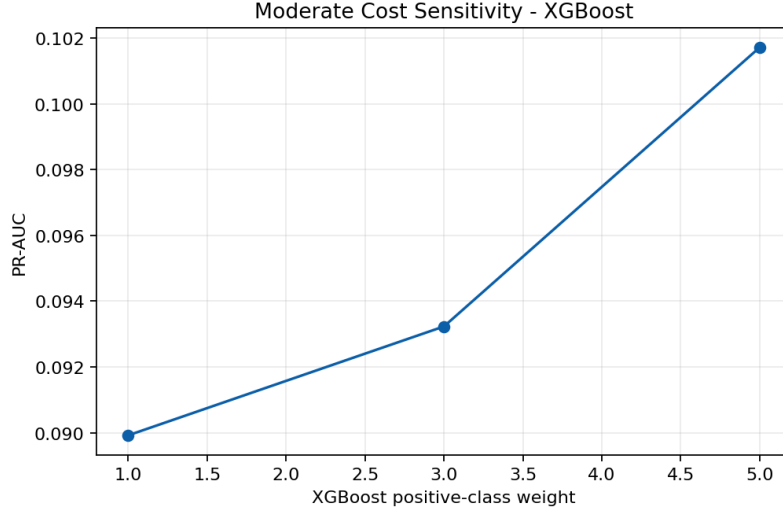
RUSBoost illustrates why recall cannot be optimized in isolation. It reaches about 61% recall but only about 3.25% precision, producing a very weak PR-AUC. The model flags many non-defaults and therefore does not provide useful risk concentration.

## 7.7 Comparison of Imbalance Treatments

**Figure 12:** PR-AUC across resampling and balanced-learning experiments

## 7.8 Cost-sensitive XGBoost

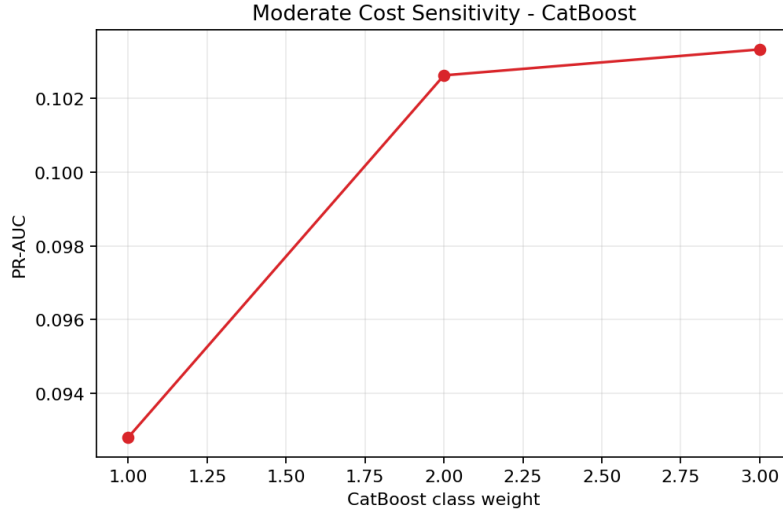
Instead of changing the dataset, cost-sensitive learning changes the loss contribution of positive observations. A series of moderate `scale_pos_weight` settings is tested.



**Figure 13:** PR-AUC under moderate XGBoost positive-class weighting

For MI-200, weight 3 produces precision around 10.98% and recall 28.16% at its Max-F1 threshold. Weight 5 produces the highest PR-AUC in this compact sweep, approximately 0.1017, but its best-F1 threshold has lower recall. The extreme theoretical imbalance ratio of approximately 69.6 was tested earlier and reduced PR-AUC to about 0.067, demonstrating that “class ratio” is not automatically the correct cost weight.

## 7.9 Cost-sensitive CatBoost



**Figure 14:** PR-AUC under CatBoost class weighting

Moderate CatBoost weights also improve minority sensitivity in some configurations. Weight 3 reaches PR-AUC about 0.1033 with precision about 11.69% and recall about 22.82% at its Max-F1 threshold.

## 7.10 Hybrid Random Undersampling + Positive Sample Weights

A final hybrid experiment tests whether class rebalancing and cost sensitivity can complement each other rather than being used separately. The training fold is first randomly undersampled so that defaults form approximately 5% of the retained sample. Positive/default observations are

then assigned a sample weight of 5 during XGBoost fitting. Thus the experiment simultaneously increases minority exposure by removing part of the majority class and increases the loss contribution of each retained default.

On the original tuning split, the strongest tested hybrid configuration is RUS to 5% positives plus positive sample weight 5. It gives ROC-AUC 0.8051, PR-AUC 0.1052 and Max-F1 0.1593, with precision 14.63% and recall 17.48%. Because this result is obtained on only 206 tuning defaults, the configuration is not accepted directly.

The same configuration is then evaluated using the five-fold stratified OOF framework on the 85% development pool. The OOF result is substantially more conservative:

**Table 4:** Hybrid undersampling + sample-weight validation

| Evaluation           | PR-AUC        | ROC-AUC       | Max-F1        | Precision | Recall        |
|----------------------|---------------|---------------|---------------|-----------|---------------|
| Single tuning split  | 0.1052        | 0.8051        | 0.1593        | 14.63%    | 17.48%        |
| 5-fold OOF           | 0.0786        | 0.8195        | 0.1372        | 12.64%    | 15.01%        |
| Current OOF ensemble | <b>0.0815</b> | <b>0.8255</b> | <b>0.1479</b> | 10.96%    | <b>22.73%</b> |

The hybrid method therefore fails to preserve its apparent tuning-set advantage under OOF validation. It produces lower PR-AUC, ROC-AUC, Max-F1 and recall than the selected ensemble. The locked holdout is not opened for this rejected candidate, preventing an additional round of holdout-driven model selection.

**Decision:** Random undersampling + positive sample weights is rejected for the final model. Although RUS 5% + weight 5 appears promising on the single tuning split, the gain does not survive five-fold OOF validation. The final model therefore continues to use the original-data MI-200 ensemble without resampling.

### 7.11 Conclusion from Imbalance Experiments

**Decision:** Aggressive balancing is rejected. Mild SMOTE is competitive, and the hybrid undersampling + sample-weight experiment also shows a temporary tuning-split gain, but neither is stable enough under OOF validation to replace the final approach. The most stable direction remains moderate cost-sensitive boosting on the original data. The data appears limited not only by minority frequency but also by class overlap and finite information in the available default cases.

## 8 Model Training

### 8.1 Logistic Regression

Logistic Regression provides an interpretable linear baseline. The predicted default probability is

$$P(Y = 1 | X) = \sigma(\beta_0 + \beta^T X) = \frac{1}{1 + \exp[-(\beta_0 + \beta^T X)]}. \quad (11)$$

Because feature scales vary widely, median imputation and standardization are used for this path. ANOVA-100 outperforms PCA-90, demonstrating that supervised feature information is more useful than purely variance-preserving components for the linear baseline.

### 8.2 XGBoost

XGBoost builds an additive ensemble of regression trees. At iteration  $t$ ,

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i). \quad (12)$$

The regularized objective is

$$\mathcal{L}^{(t)} = \sum_i l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t), \quad (13)$$

and a second-order Taylor approximation permits efficient split optimization. The project uses shallow trees, low learning rates, subsampling and column subsampling to reduce overfitting.

### 8.3 LightGBM

LightGBM is another gradient boosting framework designed for efficient histogram-based tree construction. Its leaf-wise growth can model complex interactions efficiently. In early screening, raw 300 and MI-200 representations are competitive, but LightGBM does not ultimately improve the final OOF ensemble sufficiently to justify additional ensemble complexity.

### 8.4 CatBoost

CatBoost uses ordered boosting and symmetric trees to reduce prediction shift and provide strong regularized performance. Although the dataset here is mostly numerical/anonymized, CatBoost’s tree-building strategy provides a useful source of model diversity relative to XGBoost. CatBoost MI-200 is the strongest early screening path with ROC-AUC 0.8215 and PR-AUC 0.0979.

### 8.5 Feature-path Screening Summary

**Table 5:** Selected early screening results

| Model    | Path      | ROC-AUC       | PR-AUC        | Capture@5%    |
|----------|-----------|---------------|---------------|---------------|
| Logistic | ANOVA-100 | 0.8014        | 0.0694        | 31.07%        |
| Logistic | PCA-90    | 0.7941        | 0.0602        | 28.64%        |
| XGBoost  | Raw-300   | 0.8100        | 0.0888        | 30.10%        |
| XGBoost  | MI-200    | 0.8093        | 0.0902        | 27.18%        |
| LightGBM | MI-200    | 0.8077        | 0.0845        | 29.61%        |
| CatBoost | Raw-300   | 0.8200        | 0.0898        | 27.67%        |
| CatBoost | MI-200    | <b>0.8215</b> | <b>0.0979</b> | <b>30.58%</b> |
| XGBoost  | PCA-90    | 0.7852        | 0.0690        | 25.73%        |
| LightGBM | PCA-90    | 0.7860        | 0.0610        | 25.24%        |

The screening stage narrows the search space but does not yet define the final model. The next stage addresses validation instability by generating OOF predictions on a larger development pool.

## 9 Out-of-Fold Ensemble Optimization

### 9.1 Why a Second Validation Stage Was Needed

The 15% tuning split contains only 206 defaults. A single correctly ranked default changes recall by roughly  $1/206 \approx 0.49$  percentage points. Consequently, individual tuning results are noisy, and some configurations that look strong on one split may be less stable across resamples.

After broad model exploration, the 70% training and 15% tuning partitions are merged to form an 85% development pool with 82,285 accounts and 1,166 defaults. Five-fold Stratified K-Fold cross-validation is then used to generate an out-of-fold probability for every development account.

### 9.2 Final OOF Feature Path

Mutual Information top 200 is selected as the final representation because it repeatedly performs well with nonlinear boosting, particularly CatBoost, and offers complementary signal beyond ANOVA. All final base models use the same selected feature set so that differences reflect model behaviour rather than inconsistent feature spaces.

### 9.3 Base Models

Three complementary probability models are trained in each OOF fold:

1. **XGBoost-1:** 350 trees, maximum depth 4, learning rate 0.04, minimum child weight 10, subsample 0.90, column subsample 0.85,  $\lambda = 1$ , positive-class weight 1.
2. **XGBoost-3:** identical structural settings but positive-class weight 3, providing stronger minority sensitivity.
3. **CatBoost:** 300 iterations, depth 6, learning rate 0.06 and log-loss objective.

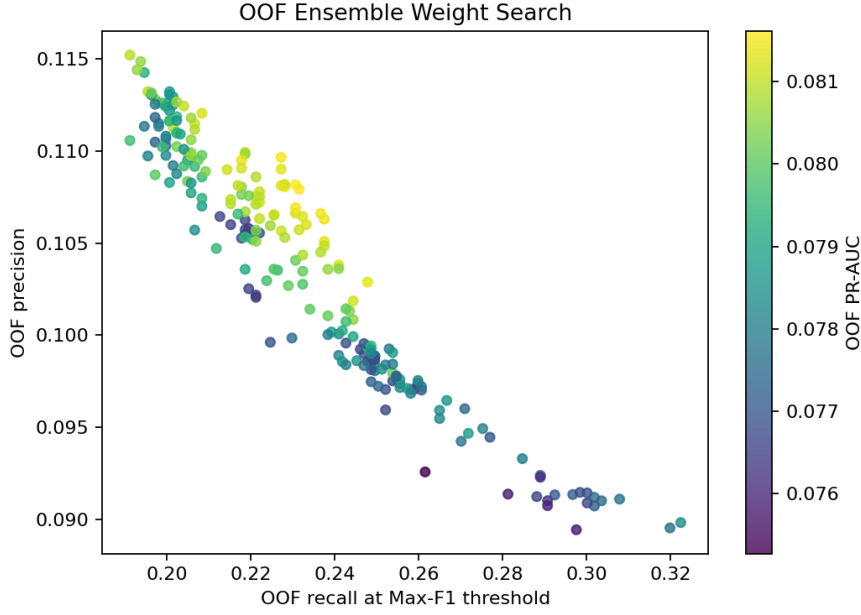
No ensemble weights are chosen before these OOF probabilities are generated.

### 9.4 Weight Search

Weights are searched over a simplex:

$$w_{x1} + w_{x3} + w_{cat} = 1, \quad w_j \geq 0, \quad (14)$$

in increments of 0.05. For each combination, OOF ROC-AUC, PR-AUC, F1, F2, precision, recall and the corresponding thresholds are recorded.



**Figure 15:** Precision-recall behaviour across OOF ensemble weight combinations

The selected F1-oriented blend is

$$0.10 XGB_{w=1} + 0.25 XGB_{w=3} + 0.65 CatBoost. \quad (15)$$

Its OOF PR-AUC is approximately 0.0815 and OOF ROC-AUC approximately 0.8255. These values are more conservative than some single tuning-split results, suggesting that OOF evaluation is successfully reducing split-specific optimism.

## 9.5 Threshold Selection from OOF Predictions

Three operating policies are designed before touching the holdout.

**Max-F1.** Choose the threshold that maximizes F1 and therefore treats precision and recall symmetrically.

**Max-F2.** Choose the threshold that maximizes F2, giving more weight to recall.

**Precision-constrained recall.** Require precision of at least 10%, then choose the threshold with maximum recall among feasible thresholds. This produces a practical recall-oriented policy without allowing precision to collapse toward the 1.42% population rate.

Once these thresholds are obtained from OOF data, they are frozen before holdout evaluation.

## 10 Final Locked Holdout Evaluation

### 10.1 Ranking Performance

The final blend is fitted on the complete 85% development pool and scored on the untouched 15% holdout.

**Table 6:** Final locked holdout ranking performance

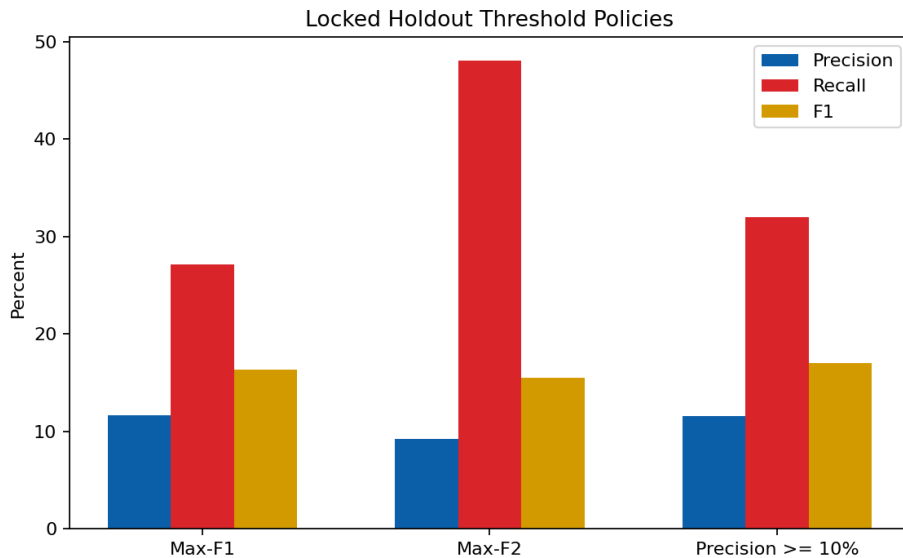
| Model                                  | ROC-AUC       | PR-AUC        |
|----------------------------------------|---------------|---------------|
| OOF-selected XGB/XGB/CatBoost ensemble | <b>0.8574</b> | <b>0.1112</b> |
| Random benchmark                       | 0.5000        | 0.0142        |

The PR-AUC is roughly 7.8 times the random prevalence baseline. ROC-AUC indicates strong overall ranking, while PR-AUC demonstrates that minority-class concentration remains materially better than chance across a range of thresholds.

### 10.2 Threshold Policies

**Table 7:** Threshold policies evaluated on the locked holdout

| Policy                    | Threshold | Precision | Recall        | F1            | F2            | Flagged |
|---------------------------|-----------|-----------|---------------|---------------|---------------|---------|
| OOF Max-F1                | 0.08466   | 11.69%    | 27.18%        | 16.35%        | 21.49%        | 3.30%   |
| OOF Max-F2                | 0.05628   | 9.25%     | <b>48.06%</b> | 15.52%        | <b>26.14%</b> | 7.37%   |
| OOF Precision $\geq 10\%$ | 0.07851   | 11.62%    | 32.04%        | <b>17.05%</b> | 23.71%        | 3.91%   |

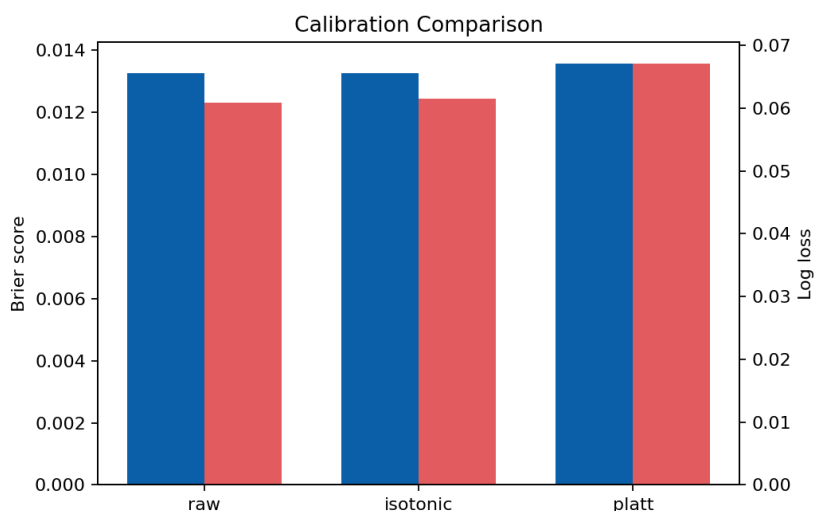


**Figure 16:** Precision, recall and F1 under the frozen OOF threshold policies

At the precision-constrained operating point, 66 of the 206 holdout defaults are detected while 502 non-defaults are flagged. The policy therefore raises recall beyond the previous low-recall region while maintaining precision above 10%.

The Max-F2 policy detects 99 of 206 defaults, nearly half of all holdout defaults, but with lower precision. This is a business trade-off rather than a modelling contradiction: lower thresholds flag more accounts and therefore increase recall while reducing precision.

### 10.3 Calibration

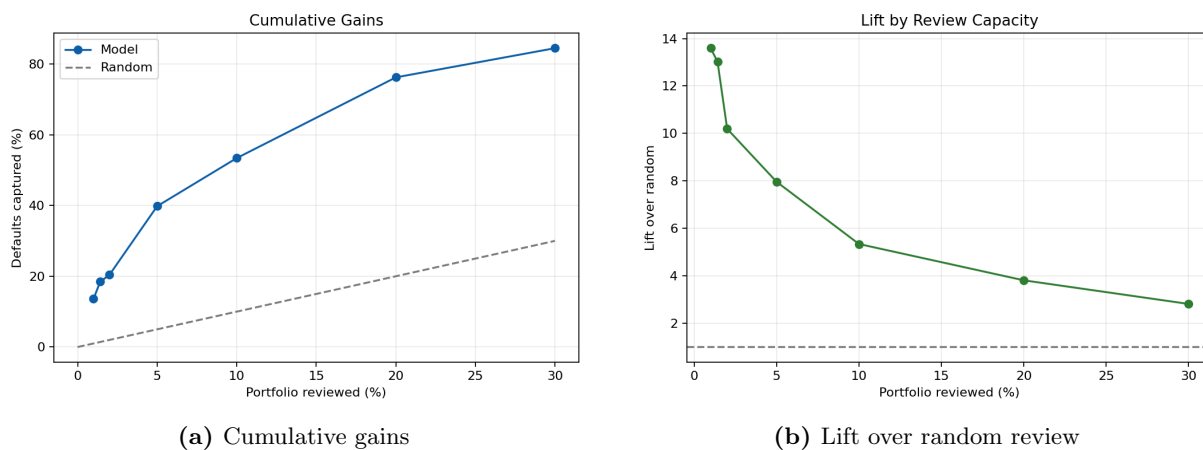


**Figure 17:** Raw, isotonic and Platt probability-calibration comparison

Raw probabilities produce Brier score 0.01326 and log loss 0.06086. Isotonic calibration slightly worsens both Brier and log loss, while Platt scaling also worsens log loss. Because calibration does not improve the executed holdout result, the raw ensemble score is retained. This preserves ranking and avoids unnecessary post-hoc complexity.

### 10.4 Capture and Lift

For credit-risk operations, ranking accounts by score can be more informative than selecting one global binary threshold.



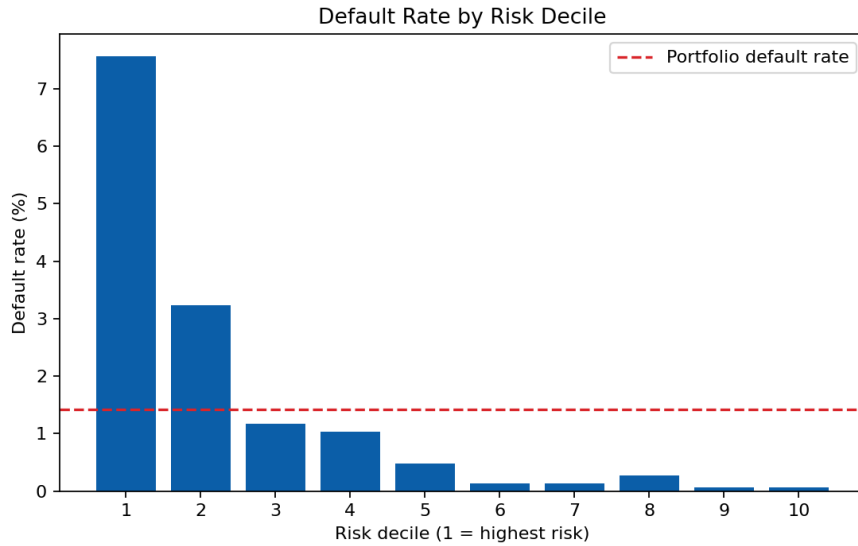
**Figure 18:** Portfolio-level ranking performance

**Table 8:** Holdout capture and lift by review capacity

| Portfolio Reviewed | Accounts | Defaults Captured | Capture | Lift   |
|--------------------|----------|-------------------|---------|--------|
| 1.00%              | 146      | 28                | 13.59%  | 13.59× |
| 1.42%              | 206      | 38                | 18.45%  | 13.02× |
| 2.00%              | 291      | 42                | 20.39%  | 10.19× |
| 5.00%              | 727      | 82                | 39.81%  | 7.96×  |
| 10.00%             | 1,453    | 110               | 53.40%  | 5.34×  |
| 20.00%             | 2,905    | 157               | 76.21%  | 3.81×  |
| 30.00%             | 4,357    | 174               | 84.47%  | 2.82×  |

The top 5% of accounts contains nearly 40% of all defaults. A random 5% review would capture only about 5%, so the model provides close to eight-fold lift in this region.

## 10.5 Risk Deciles

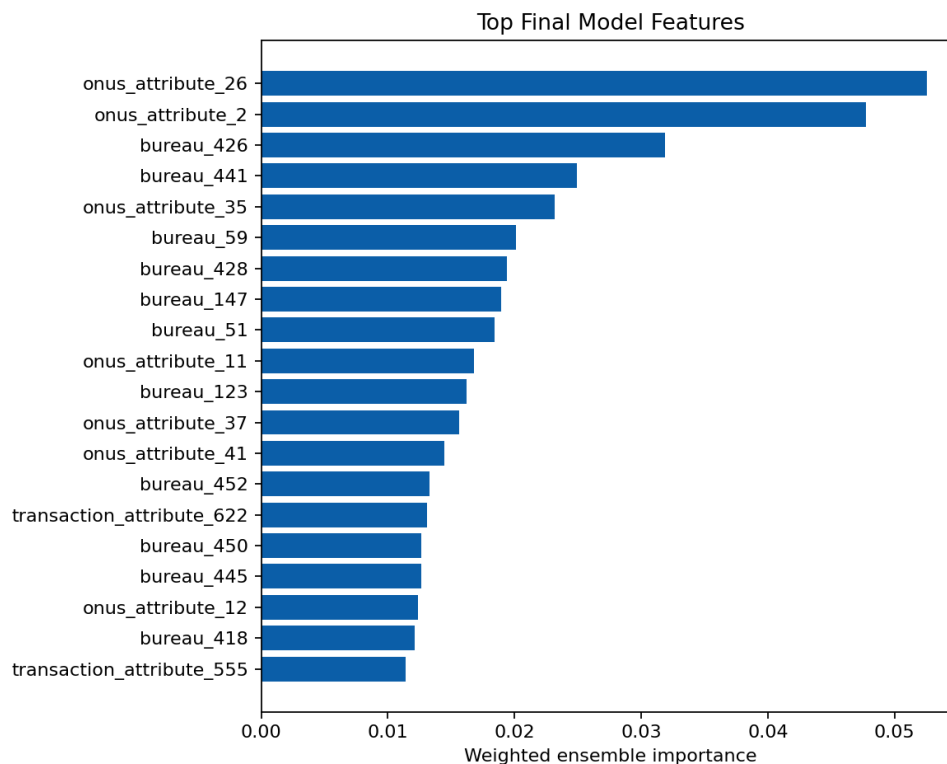
**Figure 19:** Observed default rate by predicted-risk decile

The highest-risk decile has an observed default rate of approximately 7.57%, compared with the 1.42% portfolio baseline. The first decile alone contains 53.4% of all defaults, and the first two deciles together contain 76.2%. This concentration supports the model's use as a risk-ranking instrument even though default and non-default classes overlap substantially.

## 11 Model Interpretation and Final Feature Importance

### 11.1 Weighted Ensemble Importance

The final ensemble importance is formed from normalized importance measures of the three base learners combined using the learned ensemble weights. Because the variables are anonymized, the interpretation is statistical rather than semantic.



**Figure 20:** Top features in the final ensemble

Leading variables include `onus_attribute_26`, `onus_attribute_2`, `bureau_426`, `bureau_441`, `onus_attribute_35`, `bureau_59`, `bureau_428` and other bureau/on-us attributes. This agrees with the EDA finding that on-us and bureau-related variables contain stronger standalone signal than the average transaction variable.

### 11.2 Interpretation Limitations

Because feature names do not expose their business definitions, importance should not be converted into causal statements. A high model importance indicates that a variable is frequently useful for prediction in the fitted ensemble; it does not prove that changing that variable would change default risk.

The correlation analysis also shows that several important variables are highly redundant. Importance may therefore be distributed across correlated predictors and should be interpreted at the group level where possible.

## 12 Summary Table of Experiments and Decisions

**Table 9:** Experiment log and modelling decisions

| Experiment                          | Decision           | Reason                                                                                                                      |
|-------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|
| ANOVA                               | Benchmark retained | Strong linear statistical filter, but not the best final nonlinear feature representation.                                  |
| Mutual Information top 200          | <b>Retained</b>    | Strong nonlinear selector; repeatedly competitive with CatBoost and used in final OOF stage.                                |
| PCA 90/95%                          | Rejected           | Lower PR-AUC and ROC-AUC than supervised/raw feature paths in both linear and boosted-tree comparisons.                     |
| Raw 300 features                    | Benchmark          | Competitive and useful for screening, but MI-200 offers a more compact final representation.                                |
| Logistic Regression                 | Baseline           | Provides linear benchmark and demonstrates the gain from nonlinear ensembles.                                               |
| LightGBM                            | Exploratory only   | Competitive early model, but did not add enough final OOF value to justify extra ensemble complexity.                       |
| CatBoost                            | <b>Retained</b>    | Strong individual ranking and complementary tree-building behaviour.                                                        |
| XGBoost, weight 1                   | <b>Retained</b>    | Strong clean ranking signal.                                                                                                |
| XGBoost, weight 3                   | <b>Retained</b>    | Improves minority sensitivity while remaining stable enough for OOF ensembling.                                             |
| Extreme class weight $\approx 69.6$ | Rejected           | The theoretical imbalance ratio severely reduced PR-AUC; class ratio is not automatically the correct loss weight.          |
| Random oversampling                 | Rejected           | Did not outperform the strongest boosting paths; duplicates minority examples without adding new behaviour.                 |
| Random undersampling                | Rejected           | Discards majority information and did not improve final PR-AUC.                                                             |
| RUS 5% + positive sample weight 5   | Rejected after OOF | Tuning PR-AUC rose to 0.1052, but five-fold OOF PR-AUC fell to 0.0786 with 15.01% recall; the apparent gain was not stable. |
| SMOTE 5%                            | Useful, not final  | Mild synthetic oversampling improved one XGBoost path, proving that some minority rebalancing can help.                     |
| SMOTE 10%                           | Rejected           | More aggressive synthetic balancing reduced PR-AUC materially.                                                              |
| Borderline-SMOTE                    | Rejected           | Weaker PR-AUC than cost-sensitive boosting.                                                                                 |
| ADASYN                              | Rejected           | Did not improve ranking enough to justify synthetic complexity.                                                             |
| Balanced Random Forest              | Rejected           | Lower PR-AUC and weaker threshold trade-off.                                                                                |
| EasyEnsemble                        | Rejected           | Recall improved in some operating regions, but precision/ranking remained weaker.                                           |
| RUSBoost                            | Rejected           | High recall obtained by flagging too many non-defaults; precision about 3.25% and PR-AUC about 0.025.                       |

| Experiment                      | Decision                         | Reason                                                                                            |
|---------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------|
| OOF weight search               | <b>Retained</b>                  | Learns ensemble weights from out-of-fold probabilities instead of manually choosing them.         |
| Platt calibration               | Rejected                         | Worse log loss and no ranking benefit.                                                            |
| Isotonic calibration            | Rejected                         | No useful holdout improvement over raw ensemble probabilities.                                    |
| Max-F1 threshold                | Diagnostic                       | Balanced precision/recall operating point.                                                        |
| Max-F2 threshold                | Diagnostic                       | Recall-sensitive policy detecting about 48% of holdout defaults.                                  |
| Precision $\geq 10\%$ threshold | <b>Primary operating example</b> | Maximizes OOF recall while constraining precision from collapsing toward the rare-event baseline. |

## 13 Final Model, Scoring Output, and Production Logic

### 13.1 Final Ensemble

The selected blend is

$$\hat{p} = 0.10\hat{p}_{XGB(1)} + 0.25\hat{p}_{XGB(3)} + 0.65\hat{p}_{CatBoost}. \quad (16)$$

These weights are the result of the OOF search; they are not manually fixed at the beginning of the modelling process.

### 13.2 Final Performance Summary

Table 10: Final model performance summary

| Metric                                  | Result        |
|-----------------------------------------|---------------|
| ROC-AUC                                 | <b>0.8574</b> |
| PR-AUC                                  | <b>0.1112</b> |
| Precision, precision-constrained policy | 11.62%        |
| Recall, precision-constrained policy    | 32.04%        |
| F1, precision-constrained policy        | 17.05%        |
| Max-F2 recall                           | 48.06%        |
| Max-F2 F2                               | 26.14%        |
| Capture@1.42%                           | 18.45%        |
| Capture@5%                              | 39.81%        |
| Capture@10%                             | 53.40%        |
| Capture@20%                             | 76.21%        |
| Top-decile default rate                 | 7.57%         |

### 13.3 Production Refit

Only after locked-holdout evaluation is complete are all 96,806 labelled observations used for the production refit. The same MI-selected feature path and model structures are trained on the full labelled data. This is appropriate because model choice has already been finalized; the goal is now to maximize the training information available for scoring unseen accounts.

### 13.4 Final Output File

The scoring file contains 41,792 accounts. The delivered CSV contains two columns:

- **account\_number**: unique customer/account identifier.
- **default\_probability**: continuous risk score produced by the final ensemble.

The final score distribution has mean approximately 0.0196, median 0.00921, 75th percentile 0.02284 and maximum 0.6010. Binary 0/1 predictions are not written to the output because the appropriate decision threshold depends on the bank’s review capacity and relative costs of false positives and false negatives.

### 13.5 Operational Use

A practical deployment can use the same probability score in multiple ways:

- rank accounts for manual review or collection prioritization;

- set a capacity-based top- $k$  review policy;
- use a recall-sensitive threshold when missed defaults are especially expensive;
- combine the score with business rules, exposure size and customer treatment strategies;
- recalibrate thresholds over time as default prevalence and portfolio composition change.

## 14 Key Findings, Limitations, and Future Work

### 14.1 Key Findings

- Severe class imbalance makes accuracy almost meaningless; PR-AUC, recall/precision trade-offs, capture and lift are essential.
- Missingness is structured rather than purely random. Some features have large default/non-default missingness gaps and should not be discarded simply because they are sparse.
- On-us and bureau-enquiry families contain stronger average univariate risk signal than the transaction family, although transaction attributes can contribute through interactions.
- Different feature-ranking methods identify partly different variables. Combining multiple statistical views provides a broader candidate set than choosing ANOVA alone.
- PCA is not beneficial for the final boosted-tree models because variance preservation does not align sufficiently with supervised default discrimination.
- Mild SMOTE can help, but aggressive synthetic balancing, extreme class weighting and several balanced ensembles reduce precision-recall performance.
- Combining random undersampling with positive sample weights can look strong on a single tuning split (PR-AUC 0.1052 for RUS 5% + weight 5), but the effect falls to 0.0786 PR-AUC under five-fold OOF validation, so the hybrid is rejected rather than promoted from a noisy split.
- Moderate cost-sensitive boosting offers a better compromise: minority errors receive extra attention without discarding majority information or overwhelming the learner with synthetic observations.
- OOF ensemble optimization is more reliable than selecting weights from one tuning split with only 206 defaults.
- The final model is especially strong as a ranking system: the top 5% of accounts contains about 39.8% of defaults and the top 20% contains about 76.2%.

### 14.2 Limitations

- Feature definitions are anonymized, limiting causal and domain-level interpretation.
- The number of positive examples remains relatively small. Resampling changes exposure to these cases but cannot create genuinely new real-world default behaviour.
- Holdout evaluation covers one development snapshot; temporal validation would be preferable if observation dates were available.
- The reported thresholds are illustrative operating policies. A production bank should choose thresholds using explicit monetary costs, exposure and intervention capacity.
- Calibration quality can change as portfolio prevalence changes, so ongoing monitoring is required even when raw holdout calibration is acceptable.

### 14.3 Future Work

- **Temporal Validation:** use out-of-time splits if timestamps become available to test stability across economic periods.

- **Cost-sensitive Objective Design:** optimize expected financial loss using exposure-weighted false-negative and false-positive costs rather than generic F1/F2 alone.
- **Monotonic Risk Constraints:** where business definitions of attributes become available, impose monotonic relationships for interpretability and policy consistency.
- **Advanced Stacking:** train a leakage-safe meta-model on OOF base predictions and compare it with the current weighted blend.
- **Stability Monitoring:** track feature PSI/KS, score drift, calibration drift and decile default rates after deployment.
- **Explainability:** combine tree SHAP values with feature-family aggregation once feature semantics are made available.
- **Reject Inference and Selection Bias:** if the historical dataset contains only approved credit customers, future work can account for approval-policy bias when broader applicant data becomes available.

## 15 References

### References

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